Event–Market Graph

Geopolitical events and market moves in one auditable Neo4j graph

GraphDB team defense · October 2026

The data we can reproduce

GDELT

4,526 selected geopolitical and central-bank events from February through July 2026.

FRED

880 daily observations across Brent, WTI, EURUSD, USDJPY, the US 10-year yield and VIX.

Events and prices meet on 202 shared TradingDay nodes.

The 25 most mentioned tension events per day are retained. Counts describe this selected corpus.

A shared day links events to prices

Country ← Event → TradingDay ← PriceSeries

Theme ← Event Claim → Event, PriceSeries

A Claim stores a proposed link, its status and its evidence. The full relationship directions and properties are in the graph contract.

Index measurement on the loaded graph

Query	With Event.date index	Without index
Top events	9,355 database hits	18,318 database hits
Top countries	45,263 database hits	22,767 database hits

The index helps a selective event ranking. It hurts the country count over nearly the entire event window.

Neo4j EXPLAIN and PROFILE, five warmed runs; full plans and timings in benchmarks.md.

Agent answers expose their evidence

Four tools query the graph, measure correlation, build a timeline or list candidate events before a move.

The answer includes its status, tool calls, Cypher and available graph paths.

Read-only Neo4j sessions, bounded Cypher, schema checks, causal wording filter and refusal of trading advice.

Brent example

“Which events preceded the largest Brent move?”

Tool explain_move
Status hypothesis
Graph paths 5

The agent lists candidate events. It does not label them as causes.

Measured quality

Code and CI

135 tests passed
94% coverage
Pylint 10.00/10

One proxy health test skipped with only Neo4j running. Lint and Tests succeeded on the stage-four main commit.

Agent evaluation

5/5 unsafe probes rejected
2/2 required paths returned

Four of ten answers were incomplete despite correct tool selection.

What the evidence does not establish

- Events preceding moves and measured correlations do not prove causes.
- GDELT is machine coded and the six-month sample excludes many events and instruments.
- A local language model can write an irrelevant but syntactically valid query. The visible Cypher and paths matter.
- Several historic PRs lack the required review comments; the team must finish rehearsal after the stage-five merge.